The Consumer-Relative Flip Across Domains

Preserve what the reader reads: a reconstruction–downstream dissociation across acoustic, seismic, neural, and biological signals

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Abstract

Signals are almost universally compressed to minimise reconstruction error, on the assumption that a more faithfully reconstructed signal better serves whatever reads it next. For a consumer that reads only part of what a signal carries, this assumption is directionally wrong: downstream error is governed not by the total reconstruction error $\mathrm{tr}(\Sigma_\delta)$ but by its projection onto the consumer's *read operator*, $\mathrm{tr}(P_\mathcal{C}\Sigma_\delta)$. A code that spends a fixed budget minimising reconstruction spends it largely on directions the consumer ignores; a code that instead protects the read directions reconstructs the signal *worse* yet serves the consumer *better* — a dissociation we call the **flip**. We test the flip under a single pre-registered protocol across twelve domains spanning three distinct physics: direction finding on automotive radar, microphone, and seismic arrays; the attention keys of a language model; and learned classifiers of moral judgment, music genre, and sperm-whale dialect. The flip reproduces wherever the consumer's read operator is **identifiable** — given by the geometry, or recovered blind from the consumer by probing what its output is sensitive to — and fails, informatively, where the read operator is *coupled* to the signal's own energy or the consumer does not function. We derive the failure modes as the four degeneracies of the read operator, one of which — coupling of the read subspace to the signal's own energy — is predictable before any code is built. Which code best exposes the flip tracks two axes of the representation's geometry — concentration and correlation — an exploratory diagnostic whose single-statistic shortcut we pre-register and refute. Reconstruction fidelity is the special case of a consumer that reads everything; observation, in general, is selective — and compression should be too.

Evidence tiers. Six sealed *flip* confirmations — synthetic, acoustic (LOCATA, AV16.3), seismic, retrieval, and bioacoustic consumers — carry the two-metric claim. A seventh sealed result, on LLM-attention keys, is a weaker *mechanism* confirmation (a reconstruction-tied worse-arm discrimination, not a full flip). Around them: one partial (radar, data-limited), one sealed null (gradient compression, at its tested rate), and three one-metric (A2) diagnostics (music, moral, post-RoPE keys). Every row resolves to a sealed pre-registration and a result artifact.

1 Introduction

Almost every system that compresses a signal is judged by how faithfully it can reconstruct it — mean-squared error for images, cosine similarity for embeddings, PSNR for audio. The assumption underneath is so ingrained it is rarely stated: that a code which reconstructs the signal better is a code that serves whatever reads the signal next better. For a decoder that consumes only part of what the signal carries, this assumption is not merely loose — it is *directionally wrong.*

The quantity that governs a consumer's downstream error is not the total reconstruction error $\mathrm{tr}(\Sigma_\delta)$ but its projection onto the consumer's **read operator**, $\mathrm{tr}(P_\mathcal{C}\Sigma_\delta)$ — the error in the directions the consumer actually reads. When a fixed bit budget is spent to minimise reconstruction, it is spent largely on directions the consumer ignores; a code that instead protects the read directions reconstructs the raw signal *worse* yet serves the consumer *better*. We call this dissociation the **flip**: at matched bits, read-preserving coding beats reconstruction-oriented coding on the downstream task, precisely while reconstruction-oriented coding wins on reconstruction. The two fidelities move in opposite directions.

The flip was first isolated in two narrow settings — the direction of arrival read by a sensor array, and the keys read by softmax attention in a language model — where it could be dismissed as an artifact of the geometry. Here we ask whether it is instead a **property of observation**. We test the same matched-bits dissociation, with the same discipline (each configuration frozen on a calibration split and scored on held-out data, every outcome pre-registered and reported regardless of sign), across twelve domains spanning three distinct physics — direction finding on automotive radar, hearing-aid microphone arrays, and a seismic array reading earthquake back-azimuth; the attention keys of a transformer; a fine-tuned moral-judgment classifier; music genre and, strikingly, the dialect of sperm whales in their click-rhythm codas.

The dissociation reproduces across all three physics and every consumer type wherever a single condition holds: the **read operator is identifiable** — given by the geometry, or recovered blind from the consumer by probing what its output is sensitive to. Where it is not identifiable, or where it is *coupled* to the signal's own energy (as it is for gradient compression under a curvature metric, whose high-curvature and high-energy directions coincide), the flip can be forbidden at the operating rate — and empirically vanishes. These failures are not noise; they are the theory's scope conditions. We sort them into four recurring failure modes of the operator $P_\mathcal{C}$, and show that the central one — coupling — is governed at a given budget by a finite-rate criterion, with a read–energy alignment coefficient κ as its low-rate summary (though, as we report, κ does not predict the flip's magnitude). Which encoding best exposes the flip, in turn, lives on two axes — how *concentrated* a representation's directions are and how *correlated* its channels are — and no single summary statistic (we pre-register one and refute it) can stand in for the full two-axis diagnostic.

Contributions.

1. The flip demonstrated as **domain-general** — **six** sealed *flip* confirmations spanning synthetic, acoustic, seismic, retrieval, and bioacoustic consumers, three distinct physics, plus a weaker *mechanism* confirmation on LLM-attention keys — under sealed pre-registration with outcomes reported regardless of sign.

2. A **blind read-operator probe** that recovers what a consumer reads without its ground truth, turning a failed retrieval test into a confirmation on the identical held-out set, and confirmed a second time on a fully virgin split — the identifiability boundary made constructive.

3. The failure modes derived as the **four degeneracies of** $P_\mathcal{C}$, and the coupling degeneracy given an ex-ante **sign predictor** κ that forbids the flip when the read subspace aligns with the signal's energy — together with the honest negative that κ does *not* predict the flip magnitude, which we tested and report rather than claim.

4. An **exploratory regime map** (concentration $\times$ correlation) that motivates a per-domain code selector, together with a *sealed refutation* of a single-statistic shortcut for it — the refutation, not the map, being the solid claim.

2 Theory

2.1 The controlling quantity and the flip

We take as given the second-order result established in a companion foundational work [2]. Intuitively, only the part of a coding error that falls in the directions the consumer actually reads can move its output; error in the directions it ignores is free. Formally, for a consumer $\mathcal{C}$ with output metric G acting on a representation through a Jacobian J, the leading-order downstream distortion induced by a coding error δ is

$$d_{\mathcal{O}} \;=\; \mathbb{E}_x \operatorname{tr}\big(P_{\mathcal{C}}(x)\,\Sigma_\delta(x)\big), \qquad P_{\mathcal{C}} = \mathbb{E}[J^\top G J], \tag{1}$$

reducing under stated conditions to $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$, with Σ_δ the error second moment. $P_{\mathcal{C}}$ is symmetric positive semidefinite; its range is the *read subspace* (the directions the consumer distinguishes) and its kernel is the quotient the consumer cannot tell apart. Reconstruction fidelity is the special case $P_{\mathcal{C}} = I$: a consumer that reads everything.

Fix a total bit budget and consider three codes on the representation: a *read-preserving* code (R) that allocates bits to the range of $P_{\mathcal{C}}$; a *reconstruction-oriented* code (O) that minimises $\operatorname{tr}(\Sigma_\delta)$ (reverse water-filling [3] in the signal eigenbasis); and an *anti* code (A) that starves the read directions. The **flip** is the joint event

$$\underbrace{d_{\mathcal{C}}(\mathrm{R}) \;\le\; d_{\mathcal{C}}(\mathrm{O})}_{\text{downstream: R wins}} \quad \text{and} \quad \underbrace{\operatorname{tr}(\Sigma_\delta^{\mathrm{O}}) \;\le\; \operatorname{tr}(\Sigma_\delta^{\mathrm{R}})}_{\text{reconstruction: O wins}}, \tag{2}$$

with the anti check $d_{\mathcal{C}}(\mathrm{A}) \ge \max(d_{\mathcal{C}}(\mathrm{R}), d_{\mathcal{C}}(\mathrm{O}))$. The two inequalities in (2) point in opposite directions; that is the whole content. A code can win reconstruction and lose the task.

2.2 Four observed failure modes

The flip in (2) is not automatic. It requires the reconstruction-oriented allocation to *differ* from the read-preserving one — i.e. the read subspace must be misaligned with the signal's high-energy directions at the operating rate — and it requires the read operator to be correctly known. We observe four recurring failure modes (Table 1); we do not claim they are exhaustive.

Table 1: Four observed failure modes of the flip (one fixable). These name the mode, not a defect of the true $P_{\mathcal{C}}$: D1 is a rate-dependent relation among $P_{\mathcal{C}}$, Σ_x, and R (Prop. 1); D2 is a task-validity failure; D3 is estimation error in $\hat{P}_{\mathcal{C}}$; D4 is a false invariance hypothesis.

Failure mode	Condition	Fixable?	Instance
D1 Coupling	range $P_{\mathcal{C}}$ in the MSE-active set at rate R	rate-dependent	gradient compression
D2 Precondition	read operator uninformative for the task	needs a working consumer	moral, frozen emb.
D3 Identifiability	$\hat{P}_{\mathcal{C}}$ mis-estimated	**yes** — recover by probing	legal 035→036
D4 Mechanism-absence	claimed nuisance $\notin \ker P_{\mathcal{C}}$	no — a refutation	α=1 density (NEG-11)

(D1) Coupling — read subspace inside the reconstruction code's active set. When the read subspace lies in the modes O already spends rate on (e.g. the top-energy eigenspace at a low rate), O protects the read directions and R has nothing to add. This is *budget-dependent* (Prop. 1), not a rate-independent wall; the gradient-compression null (§4) realises it at its tested rate, and §2.3 gives its low-rate summary κ.

(D2) Precondition — the read operator is task-uninformative. If the consumer's output is at chance, there is no informative read direction to protect and the flip is ill-posed. This is a task-validity failure, *not* literally $P_{\mathcal{C}} = 0$ (a broken classifier can still have a large Jacobian). Observed for moral judgment on frozen embeddings; the flip appears once the classifier is trained.

(D3) Identifiability — wrong $\hat{G}$ (hence wrong $\hat{P}_{\mathcal{C}}$). If the read operator is *estimated* rather than recovered from the consumer, the estimate can protect the wrong subspace and O wins on held-out data. This is the only fixable failure: recovering $P_{\mathcal{C}}$ from the consumer's own output sensitivity (§2.4) restores the flip.

(D4) Mechanism-absence — the claimed nuisance is not in $\ker P_{\mathcal{C}}$. If a direction hypothesised to be invariant is not actually in the consumer's kernel, protecting it buys nothing and the effect does not exist. A genuine refutation, not a rehabilitatable miss.

A demonstration shows only the wins; a theory states where it fails. The four degeneracies are what let us say, before any experiment, which domains admit the flip.

2.3 The alignment law: coupling as a continuous dial

Degeneracy (D1) is the informative one, because it is not binary. Let $\bar{P}_{\mathcal{C}}$ be the orthogonal projector onto the rank-r read subspace range $P_{\mathcal{C}}$, and let $\lambda_1 \geq \cdots \geq \lambda_d$ be the eigenvalues of the signal second moment Σ_x.

Definition 1 (read–energy alignment)**.**

$$\kappa \;=\; \frac{\operatorname{tr}(\bar{P}_{\mathcal{C}}\,\Sigma_x)}{\sum_{i=1}^{r}\lambda_i(\Sigma_x)} \;\in\; \Big[\frac{\sum_{i=d-r+1}^{d}\lambda_i}{\sum_{i=1}^{r}\lambda_i},\, 1\Big]. \tag{3}$$

κ is the fraction of the maximal r-dimensional signal energy the read subspace captures; the lower bound is attained by the bottom-r eigenspace, and $\kappa = 1$ holds iff the read subspace is *a top-r energy eigenspace (not unique under tied eigenvalues).*

Alignment alone does not decide the flip; the *budget* does. Take the Gaussian diagonal model — $\Sigma_x = \operatorname{diag}(\lambda_1 \geq \cdots \geq \lambda_d)$, read subspace the top-$r$ coordinates (so $\kappa = 1$), both codes reverse water-filling at total rate R. Even here the coupling null is exactly a *low-rate* phenomenon.

Proposition 1 (exact finite-rate coupling criterion at $\kappa = 1$)**.** *Let $R^\star = \frac{1}{2}\sum_{i=1}^{r}\log_2(\lambda_i/\lambda_{r+1})$ be the rate at which MSE water-filling just fills the read block to level λ_{r+1}. Then* (i) *for $R \leq R^\star$ the reconstruction-oriented code O activates only read modes, O and R allocate identically on the read subspace, and $\Delta = 0$ (no flip); and* (ii) *for $R > R^\star$ O spends rate on modes $> r$ that the read-preserving code R redirects to the read modes, yielding a strict flip $\Delta = d_{\mathcal{C}}(\mathrm{O}) - d_{\mathcal{C}}(\mathrm{R}) > 0$ together with $D_{\mathrm{recon}}(\mathrm{O}) < D_{\mathrm{recon}}(\mathrm{R})$. Coupling is therefore* not *a universal wall: at $\kappa = 1$ the flip is forbidden only below $R^\star$. (For $\kappa < 1$ or a misaligned operator the same active-set logic gives a phase boundary $F(P_{\mathcal{C}}, \Sigma_x, R) > 0$; we state the aligned case the experiments probe.)*

Counterexample to an unconditional null. $\Sigma_x = \operatorname{diag}(4, 1)$, read subspace the first coordinate (r=1, κ=1), $R^\star = \frac{1}{2}\log_2 4 = 1$ bit. At $R = 2$ bits water-filling activates both modes (θ_O=0.5): $d_{\mathcal{C}}(\mathrm{O}) = 0.5$, while R spends the whole budget on mode 1, $d_{\mathcal{C}}(\mathrm{R}) = 4 \cdot 2^{-4} = 0.25 < 0.5$, and O still reconstructs better ($1 < 1.25$) — a strict flip despite $\kappa = 1$. The gradient-compression result (§4) is thus one point in the no-/weak-flip region *at its tested rate*, not a rate-independent wall.

Remark (what κ does and does not predict — honest negative, with a sharpening). The natural strengthening — that Δ is *monotone* decreasing in κ, so that κ reads off the flip magnitude as a single ex-ante number — is **not supported**, and pursuing the salvage yields a precise reason rather than another failure. We built the faithful coupling model (concentrated flat-block signal energy, read dimension equal to the energy-block dimension, and the scale-free relative flip $\Delta_{\mathrm{rel}} = 1 - d_{\mathcal{C}}(\mathrm{R})/d_{\mathcal{C}}(\mathrm{O})$) and swept κ from 1 to 0. Two facts emerge. *(a) The coupling null is exact under an identifiable condition:* $\Delta_{\mathrm{rel}} \to 0$ at $\kappa = 1$ *iff* the signal support equals the read support (the background energy vanishes) — which is precisely the gradient-compression condition, where the Hessian's support coincides with the $X^\top X$ energy. *(b) In that same sharp-null regime the sweep degenerates:* once the background energy is small enough to make the null sharp, dialing κ downward moves the read subspace into the vanishing-energy background, so $\Delta_{\mathrm{rel}} \approx 0$ for all $\kappa > 0$ and jumps only at $\kappa = 0$ — the flip lands on negligible signal. A sharp null at $\kappa = 1$ and a meaningful flip at small κ are *irreconcilable* in this model class, because κ small *means* the read subspace is low-energy. κ is therefore a coupling-null predictor with an exact null condition (read support = signal support), not a magnitude dial; the flip magnitude is governed by where the read-subspace *signal* sits relative to the reconstruction allocation, which is not a single scalar. Recorded as an honest negative (`results/GO-kappa-law.json`).

The endpoints of the ordinal prediction are fixed *analytically*, not fit:

- $\kappa \to 0$ **(maximal flip).** For direction finding the read direction is $\hat{g} = P_a^\perp a'(\theta_0)$, the array-manifold derivative orthogonalised against the steering vector, which is by construction orthogonal to the dominant received energy (the steering direction). For the synthetic and planted anchors the read subspace is planted orthogonal to the signal energy. Both give $\kappa \approx 0$ and the largest observed flips.
- $\kappa \to 1$ **(null).** For gradient compression under the curvature metric the read operator is the loss Hessian $H = \frac{1}{n} X^\top \mathrm{diag}(p(1-p))X$, whose high-curvature directions coincide with the high-energy directions of $X^\top X$; hence $\kappa \approx 1$ and, as §4 confirms, $\Delta \approx 0$ while the anti-probe stays maximal — the read operator governs the task, but the flip cannot appear.

Proposition 1 makes boundary condition (D1) an ex-ante *sign* coordinate: given the read operator and the signal covariance, κ near 1 predicts *no* flip before any code is built, which is what the gradient-compression null (§4) confirms. Remark 2.3 is equally load-bearing: the attempt to promote this to a continuous magnitude law was made and *failed* the honest test, so we do not claim it. The prospective magnitude seal we had reserved (`GO-P-2026-040`) is accordingly **not sealed** — registering a magnitude prediction on an unvalidated law is exactly the over-reach the protocol exists to prevent.

2.4 Recovering the read operator

One rarely has $P_{\mathcal{C}}$ in closed form. The read operator enters three ways. *Geometric* — for sensor arrays it is $\hat{g} = P_a^\perp a'(\theta_0)$, known analytically. *Planted* — in the synthetic and LLM-rematch anchors it is constructed from a known subspace and the prediction made blind to it. *Recovered* — on real learned representations it is obtained by the **blind probe** [2]: perturb the consumer's input and read its output sensitivity. For a cosine-ranking consumer the margin sensitivity is $\partial \cos(a, b)/\partial b \approx \hat{a} - \cos(a, b)\, \hat{b}$ (the query component orthogonal to the item), and the read operator is the top-r eigenspace of $S = \langle g\, g^\top \rangle$ — low-rank for robustness, with r and the rate selected on an internal calibration split so that only a configuration that already generalises once is sealed. The blind probe is the constructive form of the identifiability face (D3): it is what makes the flip available on any real learned representation whose consumer can be queried.

3 Methods

A domain-general protocol. Every domain is tested by the same design. On the domain's native representation we build **three codes at matched total bits per sample**: read-preserving (R), reconstruction-oriented (O, Lloyd–Max or reverse water-filling), and anti (A). The consumer is the domain's own mature estimator, independent of the compressor. For each held-out instance we record the downstream error and the reconstruction error; the per-instance flip and anti checks are (2). All configuration is chosen on a calibration split and frozen before the held-out run; each domain is pre-registered (a content hash sealed and git-timestamped before any held-out instance is scored), and outcomes are recorded regardless of sign. Matched bits are audited (allocation spread $\leq$ 1–2%); flip fractions are bootstrapped over query subsets for the retrieval and classifier consumers.

The (A2) regime diagnostic. For embedding consumers we additionally run `probe_quotient` (the compression engine's calibration-time probe): at matched bits it applies a polar (angular), a per-channel (reconstruction-oriented), and a ZCA-whitened proxy code and reports the Spearman rank agreement of the declared consumer's scores under each. Two summary statistics characterise the regime: `tangential_fraction` ($\approx 1 \Rightarrow$ an angular consumer) and `median_unit_displacement` ($\sqrt{2} \Rightarrow$ isotropic directions; small $\Rightarrow$ a concentrated offset cone).

Domains and data. *Arrays* — a simulated $\lambda/2$ ULA; LOCATA [7] (8 cm nested ULA, OptiTrack ground truth); AV16.3 [14] (8-mic circular array, camera-tracked 3-D mouth); PDAR (13-element IMS seismic array, USGS catalogue back-azimuth, 34 teleseismic events; waveforms via 6); RaDICaL [16] (8-element 77 GHz FMCW virtual array). The array consumer throughout is (wideband) MUSIC [19], and the reconstruction-oriented baseline is Lloyd–Max quantization [17]. *Neural* — Llama-3.2-3B [11] attention keys (layers 8/16); a logistic-regression training step (loss Hessian as the read metric). *Learned representations* — CourtListener [10] citation pairs with LaBSE [8] (cosine ranking); ETHICS commonsense [13] with a bert-base classifier fine-tuned to 0.74 (frozen embeddings sit at chance); FMA music [4] with CLAP [24] PCA embeddings and a genre classifier; sperm-whale codas [21] (DSWP) with a clan/dialect classifier on the inter-click-interval rhythm. Held-out sets are disjoint recordings/events/opinions/ codas wherever obtainable.

Scope, operator access, and bit accounting. Three caveats bound the claims. *(i) Local surrogate.* Equation (1) is a local second-order statement for a smooth output divergence; several reported metrics (AUROC, ranking, accuracy, MUSIC peak selection) are global or non-smooth. There $P_{\mathcal{C}}$ predicts *local* score/margin sensitivity, and the experiments test whether that surrogate stays useful at finite distortion for the domain's actual metric — it does, but the implication is empirical, not exact. We keep the coupled form $\mathbb{E}_x \operatorname{tr}(P_{\mathcal{C}}(x)\Sigma_\delta(x))$; the factorised $\operatorname{tr}(\bar{P}_{\mathcal{C}}\Sigma_\delta)$ needs a decorrelation condition we do not assume. *(ii) Operator access on arrays.* The array read direction $\hat{g} = P_a^\perp a'(\theta_0)$ is the analytic manifold tangent at the source angle θ_0 — an *oracle-informed proxy*: a deployable compressor does not know θ_0, and the tangent is a local approximation to the actual wideband-MUSIC read operator (covariance estimation, denoising, eigenspace extraction), which our own LOCATA development found can differ. We therefore read the array rows as mechanism evidence that a correctly-oriented read operator produces the flip; the *non-oracle* form — $P_{\mathcal{C}}$ recovered by probing the consumer with no ground-truth angle — is what the blind-probe domains (legal, whale, moral) demonstrate, and extending it to arrays is the natural next test. *(iii) Bit accounting; operator vs. subspace.* Matched bits include all transmitted side information (norms, scales, zero-points, and any per-instance basis); a shared calibration transform costs no per-sample

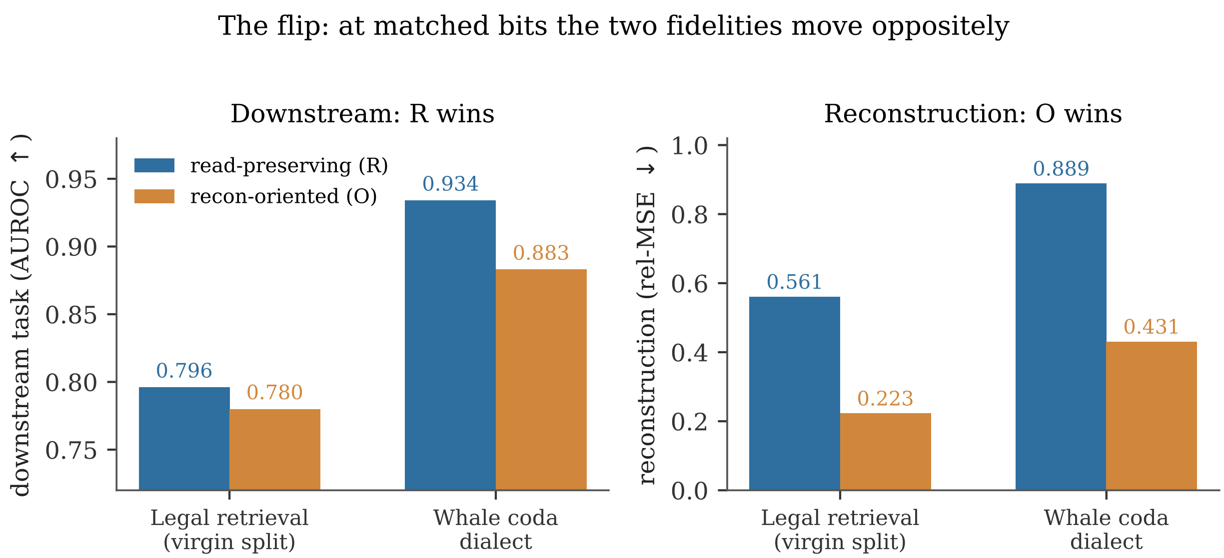

Figure 1: **The flip, on two real held-out consumers.** At matched bits the read-preserving code (R) beats the reconstruction-oriented code (O) on the downstream task (left; AUROC ↑) *while* O reconstructs the raw signal better (right; relative MSE ↓). The two fidelities move in opposite directions — a code can win reconstruction and lose the task. Numbers are held-out (legal on the virgin split, §4; whale n=2907).

bits, an instance-dependent basis does. The codes use the *projector* onto range $P_{\mathcal{C}}$; recovering the full operator with its eigenvalue weighting is strictly stronger than recovering the subspace, and we claim only the latter where only the subspace is probed. "Reconstruction-oriented" (O) means MSE-optimal *within the specified scalar allocation family* (Lloyd–Max), not the global multivariate optimum for correlated channels.

4 Results

Table 2 is the spine; each row wears its tier. Figure 1 shows the dissociation on two consumers; Figure 2 places it across the sweep.

The flip reproduces across three physics. On the physical arrays it holds under the mature subspace estimator and against independent ground truth: LOCATA **11/13** held-out recordings with the reconstruction-trade at **13/13** (Lloyd reconstructs better every time yet is downstream-worse); AV16.3 **74%** of held-out windows with recon-trade **100%** and anti-worst **76%**, against camera-tracked angle on *disjoint* recordings; PDAR seismic back-azimuth **76%** of held-out events, recon-trade **100%**, anti **76%**, against catalogue ground truth. The synthetic anchor flips 6/6 under two independent estimators; the Llama-3.2-3B key consumer shows the read-operator-projected error predicting the softmax-KL loser on **16/16** heads while exactly-tied reconstruction cannot (a reconstruction-invisible *worse-arm* discrimination, distinct from a two-consumer inversion; §7). The battery's pre-registered prediction — a flip in ≥ 2 of its three core prospective domains — is met by the two confirmed arrays.

Table 2: The twelve-domain sweep. Tiers: sealed *flip* confirmation, mechanism confirmation, partial, sealed null, (A2) diagnostic. The outcome column is descriptive; per-unit effect sizes and uncertainty are in Appendix A. Every row resolves to a sealed pre-registration and a result artifact (App. A, B).

Domain	Consumer / read operator	Held-out outcome	Tier
Synthetic ULA	beamformer + root-MUSIC / $\hat{g}$	flip 6/6 (two estimators)	flip conf.
LOCATA acoustic	wideband MUSIC / array manifold	flip 11/13, recon-trade 13/13	flip conf.
AV16.3 acoustic	wideband MUSIC / array manifold	flip 74%, recon-trade 100%	flip conf.
PDAR seismic	wideband MUSIC baz / array manifold	flip 76%, recon-trade 100%	flip conf.
Legal retrieval	cosine ranking / blind probe	R 0.796 > O 0.780 (virgin)	flip conf.
Whale codas	clan/dialect classifier / blind probe	R 0.934 > O 0.883	flip conf.
LLM keys (Llama-3.2-3B)	softmax attention / blind probe	worse-arm 16/16 (recon tied)	mechanism conf.
RaDICaL radar	MUSIC DOA / virtual array	flip 25/25, anti 60% (data-limited)	partial
Gradient optim.	curvature metric / loss Hessian	flip 27% ($<\frac{1}{2}$), anti 300/300	sealed null
Music genre	genre classifier / weight	polar 0.994 > per-ch 0.956	(A2) diag.
Moral judgment	fine-tuned classifier / weight	polar preferred (once trained)	(A2) diag.
Post-RoPE KV keys	attention logits / `a2_probe`	whitened family preferred	(A2) diag.

Recovering the read operator makes the flip transfer, twice. On real legal-citation retrieval an *estimated* read operator (document covariance) overfits: on held-out opinions reconstruction-oriented *beats* read-preserving (AUROC 0.773 vs 0.757), a registered miss (`GO-P-2026-035`, degeneracy D3). Replacing it with the blind-probe read operator — the only change — reverses the verdict on the identical held-out set (0.779 vs 0.771; `GO-P-2026-036`). Because that split had itself produced the 035 verdict, we treat 036 as *method development*, not an independent confirmation. We therefore re-ran the *frozen* recipe on a fully virgin, opinion-disjoint split (evaluation opinions never previously scored), sealed before any virgin instance was embedded (`GO-P-2026-039`): all four bars pass and more cleanly — **R 0.796 > O 0.780**, a margin double that of 036, with R edging even the uncompressed embedding (0.794, nuisance-stripping denoises the ranking), recon O 0.223 $\leq$ R 0.561, flip 200/200, anti 200/200. Moral judgment shows the consumer precondition (D2): frozen embeddings classify at chance (0.58–0.60 vs 0.63 majority) and the flip is ill-posed; a classifier fine-tuned to 0.74 makes the angular code preserve the class logit (0.994 vs 0.956 at 1 bit).

The whale flip is sealed. Sperm-whale coda dialect is the biologically most distant domain and the correlated-isotropic cell of the regime map (isotropic directions, yet correlated channels). It was promoted from an exploratory (A2) verdict to a sealed flip (`GO-P-2026-038`): with the clan classifier as consumer and the blind margin probe as read operator, on held-out disjoint codas ($n = 2907$) read-preserving AUROC **0.934** > **0.883** reconstruction-oriented, flip **300/300**, anti **300/300**, while reconstruction-oriented reconstructs the click-interval features 2× better. The map's novel axis now rests on a confirmation.

Two boundary conditions bound the claim. Gradient compression under a curvature metric is the *coupling* boundary (D1, $\kappa \approx 1$): the anti-probe is worst on **300/300** held-out steps — the Hessian read operator genuinely governs the update — yet the flip is **82/300 (27%)**, significantly *below* $\frac{1}{2}$ ($p < 10^{-3}$), not at chance: reconstruction-oriented mildly but systematically wins downstream. This is exactly what near-coupling predicts — at $\kappa \approx 0.8$ the variance-driven O allocation already captures the read-operator mass that R's top-r subspace allocation slightly misses — so at this budget coupling does not merely null the flip, it gently reverses it. No read-operator recovery can rescue it; the operator was already exact. Radar is the data limit: the flip and reconstruction-trade

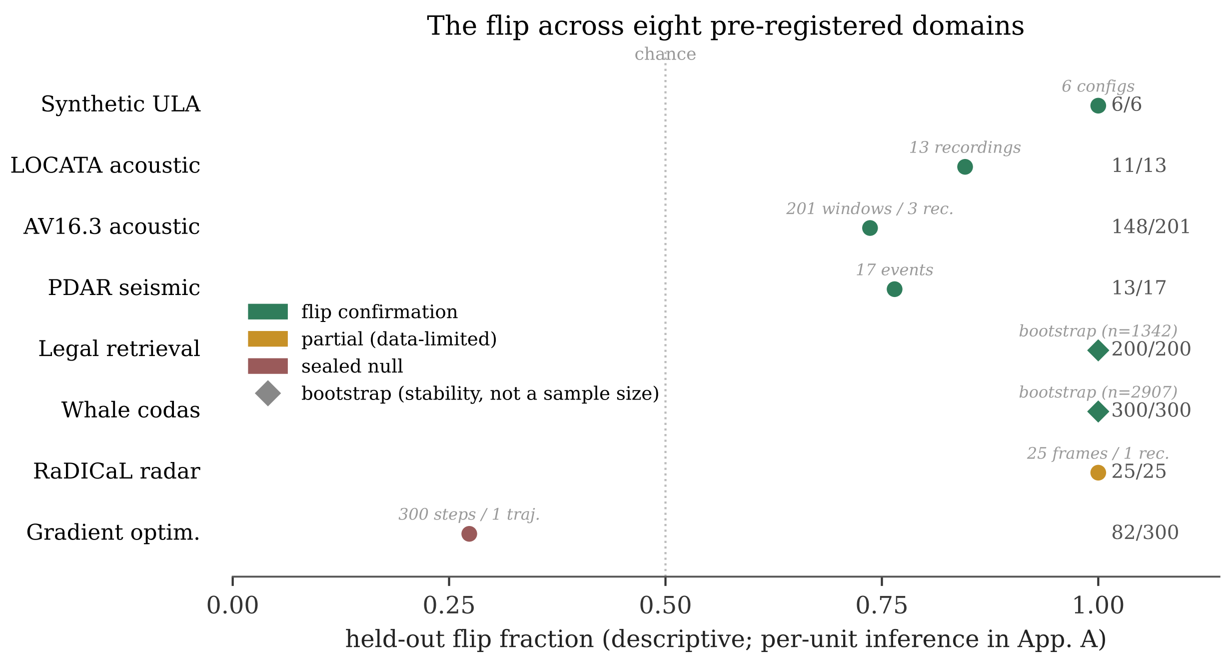

Figure 2: **The flip across eight pre-registered domains (descriptive).** Held-out flip fraction — the fraction of the domain's units (recordings, events, windows, frames, steps, or bootstrap resamples, labelled) on which R beats O — coloured by tier. *No binomial interval is shown*: the units are clustered (windows within recordings) or bootstrap resamples (legal, whale; diamonds), so a common Wilson interval has no valid interpretation. Per-unit effect sizes and the clustered/hierarchical inference are in Appendix A. Six flip confirmations span synthetic, acoustic, seismic, retrieval, and bioacoustic consumers (three distinct wave physics — acoustic, seismic, electromagnetic — plus non-physical consumers); radar is a data-limited partial; gradient compression sits *below* the $\frac{1}{2}$ line, the mild reversal that near-coupling predicts. The LLM-keys result is a mechanism confirmation, not a flip, and is not plotted.

hold **25/25** on real 77 GHz FMCW returns, but the anti control reaches only 60% on the 8-element array and the disjoint-recording held-out is inaccessible (the full set is 310 GB) — recorded as a partial, not inflated to a confirmation.

A blind, non-oracle prospective test. The sweep's confirmations recover the read operator, but each is scored after the read direction is known. We therefore ran one *fully prospective* test (`GO-P-2026-041`) on a fresh, untouched domain (20-Newsgroups `sci.space` vs `rec.autos`, frozen LSA embedding fit on the training split only): recover $P_{\mathcal{C}}$ *non-oracle* by finite-differencing the classifier's margin (no labels, no ground-truth direction), freeze the code on an internal split, and **commit the winning code, the sign of the reversal, and a magnitude band before opening the test split**. Outcome — **partial, reported in full**: the blind prediction of the *winning code* held (test AUROC R 0.975 > O 0.910, anti worst), so the framework discovered the operative geometry and chose the better code before seeing any downstream label; but the reconstruction-trade collapsed to a tie on held-out (O 0.528 ≈ R 0.526) — a mechanism confirmation, not a full flip — and the observed gap (0.064) exceeded the sealed magnitude band $[0.015, 0.045]$ (predicted sign correct, point magnitude under-estimated). The prospective *direction* is recoverable blind; the

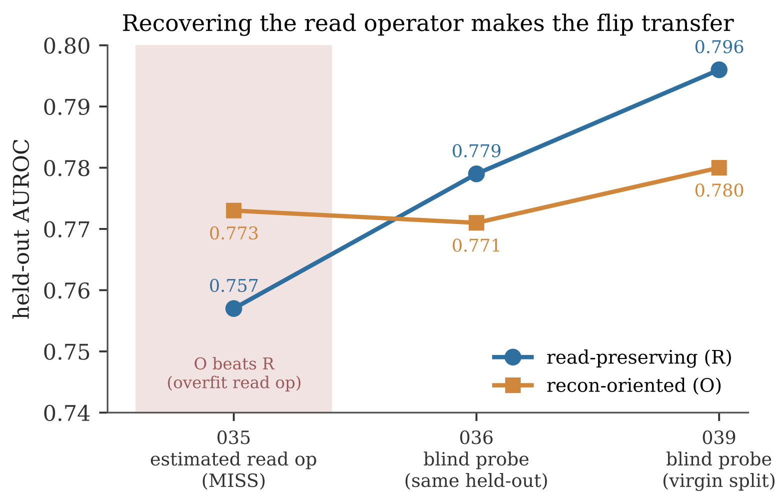

Figure 3: **Recovering the read operator makes the flip transfer.** On legal-citation retrieval an *estimated* read operator overfits and reconstruction-oriented (O) beats read-preserving (R) on held-out data (`GO-P-2026-035`, shaded miss). Replacing it with the GO-1 blind probe — the only change — reverses the verdict on the identical held-out set (`GO-P-2026-036`) and again on a fully virgin, opinion-disjoint split (`GO-P-2026-039`). The 035 miss was a read-operator-identifiability failure (D3), not a failure of the flip.

point magnitude at a single rate is not — but, as we show next, the theory does predict magnitude *across* rate.

Magnitude tracks the theory across rate. The blind test missed the point magnitude, but the functional relationship holds. On the same fresh domain we swept the bit budget and compared the theory's predicted gap $\Delta_{\text{pred}} = \text{tr}[P_{\mathcal{C}}(\Sigma_\delta^{\text{O}} - \Sigma_\delta^{\text{R}})]$ against the measured downstream gap (AUROC R − O) at each rate (Fig. 4). They track at Pearson $r = 0.995$, both decreasing monotonically toward high rate — exactly the finite-rate behaviour of Prop. 1 (the flip vanishes as R grows past $R^\star$). So Δ_{pred} predicts the effect size up to a domain-specific scale constant, not merely its sign; calibrating that constant *across domains* is the natural next contribution.

5 An exploratory regime map

This map is *exploratory*: a descriptive summary with few natural domains per cell and informally-set thresholds, presented to motivate a prospective selector study, not as a validated two-statistic predictor. Refuting a one-statistic shortcut (below) does not validate the two-axis summary; the reliable selector remains the end-to-end low-bit probe. With that caveat, characterising every embedding consumer with `probe_quotient` places them on two axes — direction concentration and cross-channel correlation — and the winning code tracks both (Table 3). Music (isotropic, low correlation) is preserved by the generic angular code and reconciles a pre-existing 6σ music sign-flip; legal (concentrated, channel-structured) needs the per-channel or blind-probe code; post-RoPE keys and sperm-whale codas both need the whitened code, but for different reasons — the keys are concentrated-correlated, the codas isotropic-correlated (their inter-click intervals are temporally correlated). Whale is the decisive case: isotropic like music yet whitening-dependent like the keys, so a single concentration statistic cannot predict it. Indeed a pre-registered `unit_displacement`

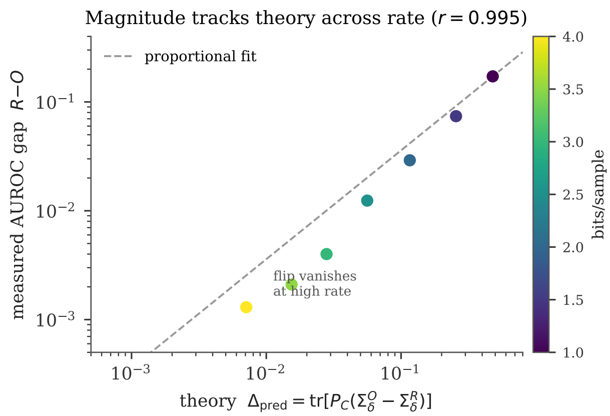

Figure 4: **Magnitude, not just sign.** On the fresh blind-test domain, the theory's predicted gap Δ_{pred} and the measured held-out AUROC gap track across bit rate at $r = 0.995$ (log–log; dashed = proportional fit). Both shrink toward high rate as Prop. 1 requires. The prediction is the effect size up to a scale constant.

threshold is *refuted* on its first out-of-sample test (`GO-P-2026-037`, ledger NEG-14): synthetic concentrated embeddings sit below the threshold yet the generic polar code wins. The reliable diagnostic is the full end-to-end probe run per domain at low bits.

Table 3: The exploratory regime map. Cells give the winning code; the empty cell (concentrated / low-correlation) is *constructible but unpopulated by natural data* in our sweep (App. A). The row labels are three correlation regimes, so "two axes" is a simplification.

	isotropic (unit_disp $\approx \sqrt{2}$)	concentrated (unit_disp small)
low correlation	music, moral $\rightarrow$ **polar**	*(structurally rare)*
cross-channel correlated	**whale** $\rightarrow$ **whitened**	keys $\rightarrow$ **whitened**
channel-structured	—	legal $\rightarrow$ **per-channel / blind-probe**

6 Related work

The pieces of this paper are individually classical; the contribution is the synthesis and its test. **Loss-aware compression as a metric.** Weighting quantization error by a curvature operator is the Hessian-pruning line — Optimal Brain Damage and Surgeon [12, 15] through GPTQ [9] — where the second-order loss sensitivity says which parameters to coarsen. Our $P_{\mathcal{C}}$ is that operator made general: not the loss Hessian of one task but any consumer's output-metric pullback $J^\top GJ$, and, critically, *recoverable from black-box queries* (§2.4) rather than requiring the loss in hand. **Remote source coding.** The coding problem — compress X to serve a downstream functional, not to reconstruct X — is the indirect/remote source-coding setting of Dobrushin and Tsybakov [5] and Wolf and Ziv [23]; the weighted-quadratic fidelity $\text{tr}(P_{\mathcal{C}}\Sigma_\delta)$ is its second-order form, and $P_{\mathcal{C}} = I$

recovers Shannon reconstruction [3, 20]. Our addition is the middle term as a *geometric, recoverable, water-fillable* operator and the prospective, cross-domain test of the resulting flip. **Information bottleneck.** IB [22] optimises a *stochastic* encoding to trade compression against relevance to a target variable; $P_{\mathcal{C}}$ is instead a *fixed geometric* operator recovered by probing a given consumer, with no relevance variable optimised — the two are not contained in one another. The components of $P_{\mathcal{C}}$ as a pullback metric are classical information geometry [1]. **Perceptual audio coding** is the historical anchor: MP3's psychoacoustic masking model [18] is precisely a hand-built read operator for the human auditory consumer, discarding what the ear cannot hear. The flip is the general statement of that idea — a read operator *recovered by probing arbitrary consumers*, not designed once for one physiology, and validated by a sealed reconstruction-vs-downstream dissociation rather than assumed. The deltas over all of the above are the same three: blind, validated read-operator recovery; six prospective flip confirmations across three physics; and the failure-mode taxonomy that says, ex ante, where the flip is unlikely to appear.

7 Discussion

What is established, and at what tier. The core result is **six sealed flip confirmations** — synthetic, LOCATA, AV16.3, PDAR, legal (confirmed on a virgin split, §4), and whale — each a two-metric dissociation scored on held-out data under a pre-registered protocol. A seventh sealed result, on LLM-attention keys, is a *mechanism* confirmation, not a full flip (below). Around them: one **partial** (radar, data-limited), one **sealed null** (gradient compression, at its tested rate), and three one-metric **(A2) diagnostics** (music, moral, post-RoPE keys). The same registration machinery that seals a confirmation also makes an exploratory row legible as exploratory; the full registry accounting is Appendix B.

Three tiers of evidence, kept distinct. We reserve *flip* for the two-metric dissociation (2). A **mechanism confirmation** is weaker: the read-operator-projected error discriminates two reconstruction-*tied* arms on a single consumer (the LLM-keys result), showing that $P_{\mathcal{C}}$ governs the task but not the two-metric trade. An **(A2) diagnostic** is weaker still — which quantizer family best preserves the consumer's rank ordering at matched bits, a one-metric verdict. Six rows are flips, one is a mechanism confirmation, three are (A2) diagnostics; we do not pool them into a single success count.

The failure modes bound the claim. The four observed failure modes (§2.2) say where the flip does not appear: identifiability is fixable by recovery; coupling is budget-dependent (Prop. 1); precondition needs a working consumer; mechanism-absence is a refutation. The gradient-compression null realises coupling at its tested rate. The refuted single-statistic regime shortcut is the same discipline applied to the map: a predictor pre-registered and killed on its first out-of-sample test.

The information-theoretic frame, and convergent evidence. That reconstruction fidelity is the special case $P_{\mathcal{C}} = I$ places the flip inside rate–distortion theory [3, 20] as its consumer-relative generalisation [2]. Independent arrival at the same object is telling: a bioacoustics decoder-robustness index — the public name for the read- operator-distortion measure whose provenance we footnote[1] — measures precisely read-operator distortion (invariant acoustic transforms leave the

[1] Originally the "Bond Index" in the ethics setting; renamed the Decoder Robustness Index (DRI) as the public index name in the bioacoustic setting.

decoder's read unchanged; stress transforms move it), i.e. the same methodology reached from a different field.

Limitations, and the open problem. The three (A2) diagnostics (music, moral, keys) are one-metric, not sealed flips; the array read operators are oracle-informed proxies (Methods); radar is data-limited; each domain uses a single model and dataset; AV16.3 now carries its recording-clustered CI (App. A) but an independent replication remains open. On magnitude: the blind non-oracle test (`GO-P-2026-041`) recovered the winning code's *direction* on untouched data but missed the sealed point magnitude at its single rate. Magnitude is nonetheless predictable *across* rate — Δ_{pred} tracks the measured gap at $r = 0.995$ (Fig. 4) — so what remains is calibrating the cross-*domain* scale constant hierarchically, the natural next contribution. And the κ coordinate delivers less than we first hoped: it robustly predicts the coupling *null* ($\kappa \to 1 \Rightarrow$ no flip) but not the flip magnitude (Remark 2.3), a limitation we found by testing the stronger claim and reporting its failure rather than the claim.

Outlook. The operational reading is direct: for compression under a known consumer, *measure the regime and adapt the code* — certify what the consumer reads rather than promise reconstruction. There is no universal code; there is a diagnostic, on two axes, and a selector that reads it — and, at the coupling boundary, a finite-rate criterion (Prop. 1) that says whether, at a given budget, the flip can appear at all.

Reproducibility statement

All code, sealed pre-registrations, and result artifacts are public in the program repository (`https://github.com/ahb-sjsu/geometric-observation`) and the companion Zenodo record (doi:10.5281/zenodo.21423315): each domain's harness, its frozen configuration and content-hash seal, the held-out result JSON, and the figure-generation script (every figure regenerates from the sealed result JSONs). The registration content hashes and the seal-before-run commit ordering (Appendix B) are verifiable in the git history. No held-out instance was scored before its registration was committed.

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A Per-domain detail

Table 4 reports each row at its *genuine independent unit* — the recording, event, window, opinion, coda, model, or seed — with the held-out flip fraction and, where the read metric is AUROC, the downstream effect size. **We do not report unified binomial p-values.** The flip fraction is a descriptive summary; for legal and whale it is computed over *bootstrap resamples of the same held-out set*, which are dependent, so we report it as a *sign-stability* diagnostic, not a sample size — and we withdraw the earlier $p < 2^{-200}/2^{-300}$ claims. AV16.3's 201 windows are clustered within 3 recordings; we ran the recording-clustered (hierarchical) bootstrap — resample the 3 recordings, then windows within each — and the flip fraction's 95% CI is $[0.68, 0.88]$, still excluding chance, versus a too-narrow naive window CI of $[0.70, 0.82]$; the raw paired angular difference is near-parity (median $\Delta_i = L_i(\mathrm{O}) - L_i(\mathrm{R})$ CI $[-0.49, +0.09]^\circ$), so AV16.3's two-metric flip rests on the tolerance-level win plus the recon-trade, not a large raw angular gap. The anti arm is a mechanistic control (deliberately non-exchangeable), so we report its fraction rather than a worst-of-three chance test.

The empty regime cell, filled by construction. The concentrated / low-correlation cell of the regime map (Table 3) is not merely argued empty: the `GO-P-2026-037` refutation instances — synthetic concentrated embeddings with a per-dimension offset and little cross-channel correlation — *are* that cell, and there generic polar wins (which is why the single-statistic shortcut, predicting "needs blind-probe" from concentration alone, was refuted). So the cell is *constructible* (we built instances in it) yet *unpopulated by natural data* in our sweep: every real embedding-consumer we

Table 4: Per-domain held-out summary at the genuine independent unit. Fractions are *descriptive*; bootstrap rows (legal, whale) are sign-stability diagnostics, not sample sizes. Effect = held-out AUROC R − O or the MSE recon-trade direction.

Domain (reg.)	unit (n)	flip fraction	downstream effect / recon-trade
Synthetic ULA (031)	configs (6)	6/6	2 independent estimators; recon-trade holds
LOCATA (033)	recordings (13)	11/13	recon-trade 13/13 (Lloyd better, worse DOA)
AV16.3 (034 D2)	windows (201; 3 rec.)	148/201	recon-trade 201/201; clustered flip CI [.68, .88]
PDAR (034 D3)	events (17)	13/17	recon-trade 17/17
Legal (039)	bootstrap (n=1342)	200/200 (stab.)	AUROC R−O = +0.016; recon $O \leq R$
Whale (038)	bootstrap (n=2907)	300/300 (stab.)	AUROC R−O = +0.051; recon $O \leq R$
LLM keys (021)	heads (16)	—	mechanism: worse-arm 16/16, recon tied
Radar (034 D1)	frames (25; 1 rec.)	25/25	recon-trade 25/25; anti 15/25 (partial)
Gradient (034 D4)	steps (300; 1 traj.)	82/300	below $\frac{1}{2}$: O mildly wins; anti 300/300
Moral (037)	(A2) diagnostic: polar 0.994 > per-ch 0.956 at 1 bit (one metric)		
Music (037)	(A2) diagnostic: reconciles the 6σ music sign-flip; polar		
KV keys (037)	(A2) diagnostic: whitened family preferred at matched bits		

measured that was concentrated had also acquired cross-channel correlation through its shared offset component. Constructible-but-naturally-empty is a stronger, data-backed statement than a genericity claim.

B Registry accounting

The "no file drawer" claim rests on the registration sequence having no unexplained holes. Table 5 gives every `GO-P-2026` identifier (001–041) a disposition: *sweep* (a row of Table 2), *core* (a falsifiable-core row of a companion paper), *cost* (a companion theorem harness), *sup→NNN* (a miss superseded by a named successor, the miss still reported), or *none* (never assigned). The two skips, 029 and 030, carried no file and no run at any commit (verified absent from the full git history). Every sealed row's registration is a strict git ancestor of its result commit (checked by `merge-base`), on GPG-signed commits; supersession chains 001→002, 007→009, 012→014, 016→019, 020→021, 032→033, 035→036→039 carry each miss at equal prominence with its correction. Two honest limits: git ancestry establishes ordering *within the repository*, not immutable wall-clock time (that would need external timestamping beyond the signed history); and *dataset-level* freshness — a test set never inspected under a superseded registration — is met explicitly only by the virgin legal split (`GO-P-2026-039`), the other confirmations registering a new method or configuration before its run but on previously-seen data distributions.

C The κ measurement (`results/GO-kappa-law.json`)

We computed κ (Definition 1) for the two analytic anchors ($\kappa \approx 0.006$ for the synthetic-ULA read $\hat{g} = P_a^{\perp} a'$; $\kappa \approx 0.80$ for the gradient Hessian on `load_digits`), and ran a controlled synthetic sweep dialing κ from 0 to 1 with real matched-bit quantization to test whether the flip magnitude Δ is monotone in κ. It is not: Δ is peaked rather than monotone (no tested normalization — Δ, $\Delta / \operatorname{tr}(\bar{P}_{\mathcal{C}} \Sigma_x)$, or $d_{\mathcal{C}}(\mathrm{O})/d_{\mathcal{C}}(\mathrm{R})$ — is monotone), because a low-energy read subspace carries little signal and because the result depends on the quantizer-range regime (Remark 2.3). The *ordinal* content survives — the $\kappa \approx 0.80$ gradient anchor is the empirical null (flip 27%), the $\kappa \approx 0.006$ array read is a large flip. The faithful coupling salvage (Remark 2.3) then pins the null to an exact

Table 5: Disposition of every registration ID. NEG-k marks a sealed refutation carried in the ledger's Honest Negatives.

ID	disposition	ID	disposition
001	core; sup→002 (NEG-5)	021	**sweep** A2 (rematch, hit)
002	core (GO-2 neg; NEG-6)	022	cost (two-observer)
003	core (NEG-7)	023	cost (rate region)
004	core (NEG-8)	024	cost (observer mismatch)
005	core (NEG-9)	025	cost (dispersion)
006	core (GO-2 Instance 1)	026	cost; sup→027 (NEG-13)
007	core; sup→008 (NEG-10)	027	cost (floor, resolved)
008	core; sup→009	028	core (GO-6, App. B)
009	core (GO-2 Instance 2)	029	*none* (skip)
010	core (Gate B, gradient)	030	*none* (skip)
011	core (GO-1 blind probe)	031	**sweep** A1 (DOA sim)
012	core; sup→013	032	**sweep** A3; sup→033 (miss)
013	core; sup→014	033	**sweep** A3 (LOCATA hit)
014	core (GO-3)	034	**sweep** D1–D4 (battery)
015	core (GO-4)	035	**sweep** L; sup→036 (miss)
016	core; sup→017	036	**sweep** L (legal hit)
017	core; sup→018	037	**sweep** M/K/Mu (NEG-14)
018	core; sup→019	038	**sweep** W (whale)
019	core (GO-5, refuted NEG-11)	039	**sweep** L (virgin split)
020	core; sup→021 (NEG-12)	040	*void* (κ-magnitude, unsealed)
		041	**sweep** blind non-oracle (partial)

condition (read support = signal support) and shows why no magnitude dial exists in this model class. The prospective magnitude seal `GO-P-2026-040` was **not** taken: the law it would have tested did not survive validation. A magnitude law matched to the quantizer regime remains open.